Protocol Analysis Sheet / Test Log

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Webinar 4 — I2C/SPI

Environment

Field	Value
MCU	ESP32 DevKit
Peripheral	MPU6050 / 0x68
SDA/SCL	GPIO21 / GPIO22
Capture	i2c-capture.vcd
Build	BUGGY

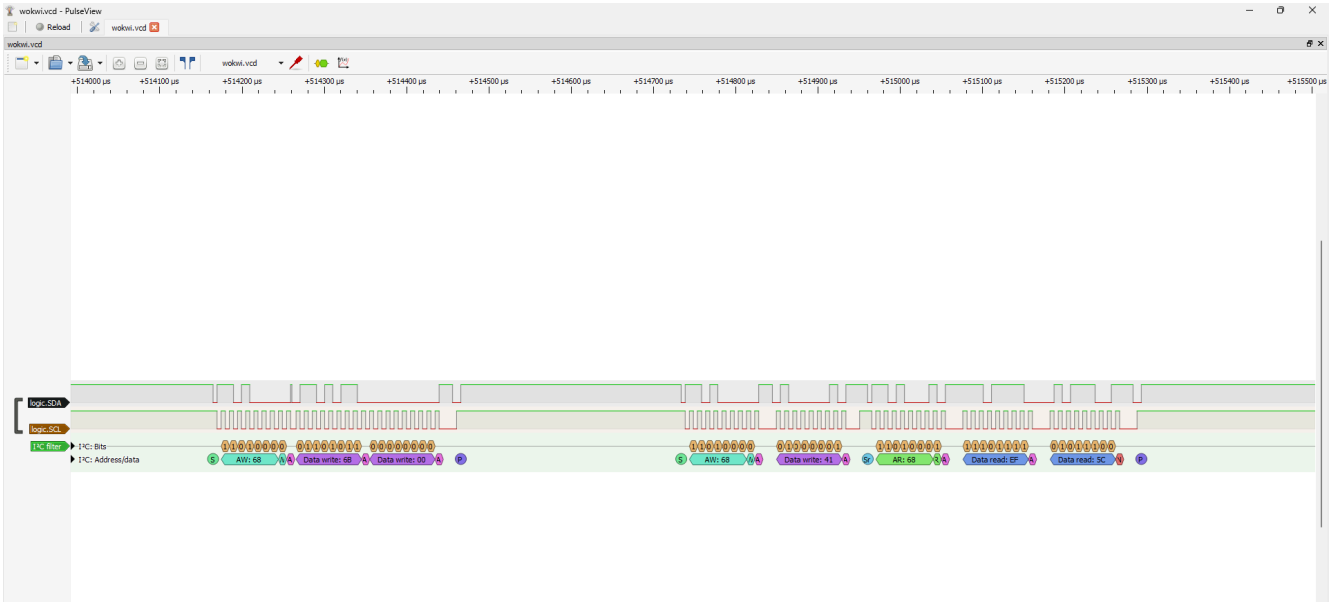
Transaction Log

#	Timestamp	Address	R/W	Data	ACK/NACK	Expected	Observed
1	+0.514164858 s +0.514733048 s	0x68	W→R	6B 00 41 → EF 5C	ACK, ACK, ACK; ACK, ACK, ACK, ACK, NACK	ACK	ACK
2	+1.515475508 s	0x68	W→R	41 → EF 5C	ACK, ACK, ACK, ACK, NACK	ACK	ACK
3	+2.516475108 s	0x68	W→R	41 → EF 5C	ACK, ACK, ACK, ACK, NACK	ACK	ACK
4	+3.517474756 s	0x68	W→R	41 → EF 5C	ACK, ACK, ACK, ACK, NACK	ACK	ACK
5	+4.518476368 s	0x68	W→R	41 → EF 5C	ACK, ACK, ACK, ACK, NACK	ACK	ACK
6	+5.519475374 s	0x68	W→R	41 → EF 5C	ACK, ACK, ACK, ACK, NACK	ACK	ACK
7	+6.520475516 s	0x68	W→R	41 → EF 5C	ACK, ACK, ACK, ACK, NACK	ACK	ACK
8	+7.521476020 s	0x68	W→R	41 → EF 5C	ACK, ACK, ACK, ACK, NACK	ACK	ACK
9	+8.522474244 s	0x68	W→R	41 → EF 5C	ACK, ACK, ACK, ACK, NACK	ACK	ACK
10	+9.523475020 s	0x68	W→R	41 → EF 5C	ACK, ACK, ACK, ACK, NACK	ACK	ACK
11	+10.524474958 s	0x69	W	-	NACK	ACK / device response	NACK

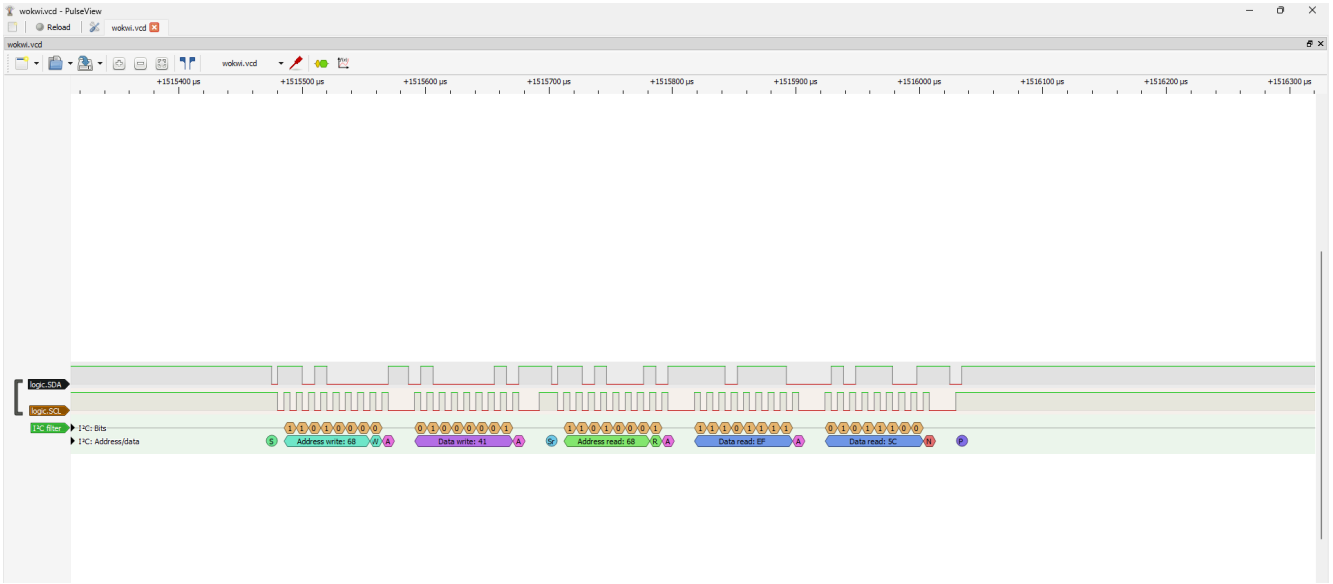
Fault Timeline

Event	Timestamp	Trace evidence	Firmware evidence
Last successful ACK	+9.523475020 s	Address Write 68 → ACK; Data Write 41 → ACK; Address Read 68 → ACK; Data Read EF → ACK Data Read 5C → NACK	[DATA] Temperature: 24.00 C
First NACK	+10.524474958 s	Address Write 69 → NACK → STOP	[E][Wire.cpp:499] requestFrom(): i2cWriteReadNonStop returned Error -1
Retry #1	-	No subsequent I2C transaction observed	I2C request returned Error -1
Retry limit / error	+10.524474958 s	Failed I2C transaction / NACK	[10524][E][Wire.cpp:499] requestFrom(): i2cWriteReadNonStop returned Error -1
Recovery ACK	-	No recovery ACK observed	Temperature did not resume / no successful I2C response observed

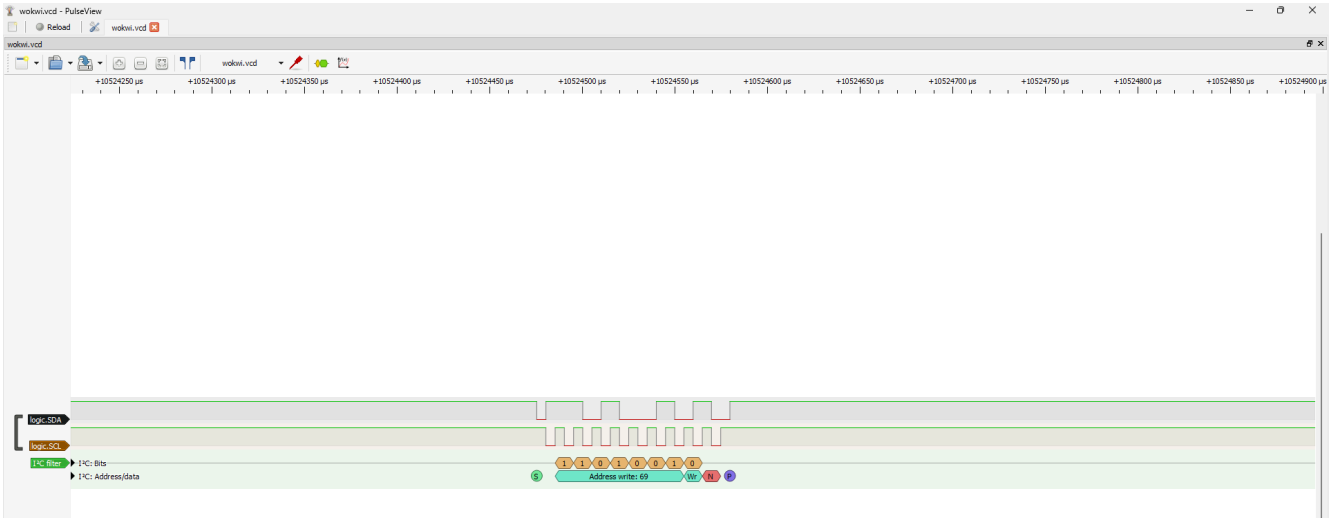
1st Transaction



2nd-10th Transaction



11th Transaction




```
Problems  Output  Debug Console  Terminal  Ports

ets Jul 29 2019 12:21:46

rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:2
load:0x3fff0030,len:1156
load:0x40078000,len:11456
ho 0 tail 12 room 4
load:0x40080400,len:2972
entry 0x400805dc

=== WEBINAR 4: I2C PRACTICE / BUGGY ===
Find the defect using trace + firmware behavior.
FAULT OFF = normal sensor; FAULT ON = simulated peripheral loss.

[DATA] Temperature: 24.00 C
                          [SYSTEM] Main loop alive
[DATA] Temperature: 24.00 C
                          [SYSTEM] Main loop alive
[DATA] Temperature: 24.00 C
                          [SYSTEM] Main loop alive
[DATA] Temperature: 24.00 C
                          [SYSTEM] Main loop alive
[DATA] Temperature: 24.00 C
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[DATA] Temperature: 24.00 C
                          [SYSTEM] Main loop alive
[DATA] Temperature: 24.00 C
                          [SYSTEM] Main loop alive
[DATA] Temperature: 24.00 C
                          [SYSTEM] Main loop alive
[ 10524][E][Wire.cpp:499] requestFrom(): i2cWriteReadNonStop returned Error -1
```

Results table

State	UART	I2C ACK/NACK	Main loop	Conclusion
FAULT OFF — normal operation	[DATA] Temperature: 24.00 C	Successful transfer, ACK	[SYSTEM] Main loop alive continues	The system works normally
FAULT ON — error	[10524][E][Wire.cpp: 499] requestFrom(): i2cWriteReadNonSt op returned Error -1	First NACK: Address Write 69 → NACK → STOP	Does not continue after the error	I2C communication lost; the system does not continue normal operation
FAULT OFF — after recovery	No new [DATA] Temperature	No recovery ACK is observed	[SYSTEM] Main loop alive does not appear	No automatic recovery; restart required

Bug Report

Title: I2C communication does not recover after FAULT

Environment: ESP32 DevKit with Peripheral MPU6050 / 0x68

Preconditions: simulator is started

Steps:

1. Start the system with FAULT OFF.

2. Observe normal I2C communication with the MPU6050.

3. Verify that UART periodically displays:

● [DATA] Temperature: 24.00 C

● [SYSTEM] Main loop alive

4. Turn FAULT ON to simulate loss of the I2C peripheral.

5.Observe the I2C traffic in PulseView and the UART output.

6. Turn FAULT OFF without restarting the system.

7. Observe whether I2C communication and the main loop recover.

Expected: After the I2C device becomes available again, the system should recover communication automatically, resume receiving temperature data, and continue the main loop without requiring a restart.

Actual:

After FAULT ON, the I2C transaction ends with a NACK: Address Write 69 → NACK → STOP

UART reports: requestFrom(): i2cWriteReadNonStop returned Error -1

After this error, [DATA] Temperature and [SYSTEM] Main loop alive messages stop.

After FAULT OFF, the system does not recover automatically. No successful I2C communication or temperature data is observed. A restart is required to restore normal operation.

Trace Evidence:

Transactions before the fault complete successfully with ACKs.

The first observed failed transaction contains:

Address Write 69 → NACK → STOP

No recovery ACK is observed after FAULT OFF

Firmware Log Evidence:

Normal operation: [DATA] Temperature: 24.00 C

Normal operation: [SYSTEM] Main loop alive

Error: requestFrom(): i2cWriteReadNonStop returned Error -1

Severity: Critical

Reproducibility: 5/5

QA Conclusion

Before FAULT occurs, I2C communication works normally: the device responds to requests, temperature data is received, and the main loop keeps running.

After FAULT is turned on, a NACK and the I2C error Error -1 appear. After that, temperature reception and the [SYSTEM] Main loop alive message stop.

In the last transaction, 0x69 is sent instead of the expected address 0x68, after which the device returns NACK.

After FAULT is turned off, no automatic recovery occurs. A restart is required to restore system operation.

Conclusion: the system handles the loss of an I2C device incorrectly and has no proper mechanism for automatically restoring I2C communication after the error is removed.